

Lab 11

1. Function Templates (Basic)

Question:

Write a **function template** findMax that returns the maximum of two values. Test it with int, float, and char.

Sample Output:

Maximum of 10 and 20 is 20
Maximum of 2.5 and 3.7 is 3.7
Maximum of 'A' and 'Z' is Z

2. Overloading Function Templates

Question:

Write **two overloaded function templates** named add:

- One takes **two arguments** and returns their sum.
- Another takes **three arguments** and returns their sum.

Sample Output:

Sum of 10 and 20: 30
Sum of 1, 2, 3: 6
Sum of 2.5 and 3.5: 6
Sum of 1.1, 2.2, 3.3: 6.6

3. Class Templates (Basic)

Question:

Write a **class template** Box<T> that stores a value of type T. Include setValue(T) and getValue(). Test it with int and string.

Sample Output:

Box holds: 100

Box holds: Hello

4. Default Arguments in Function Templates

Question:

Write a **function template** multiply with a default argument factor = 2. The function should multiply the given number by the factor.

Sample Output:

Multiply 5: 10
Multiply 5 with factor 3: 15
Multiply 2.5: 5
Multiply 2.5 with factor 4: 10

5. Default Arguments in Class Templates

Question:

Write a **class template** Storage<T = int> that stores a value. Include a function display(). Test it with and without specifying the type.

Sample Output:

Default Storage holds: 50
Storage<double> holds: 3.14